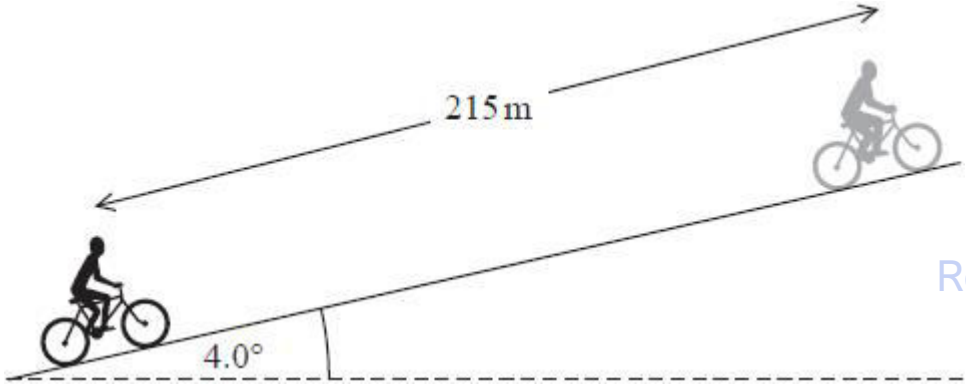
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Questions

Q1.

A cyclist is moving up a slope at a constant speed. The angle of the slope to the horizontal is 4.0°, as shown.



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As the cyclist pedals, there is a constant forward force on the bicycle of 150 N. The cyclist travels 215 m along the slope.

(a) Show that the work done by the cyclist is about 3×10^4 J.

(2)

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(b) The cyclist must do work against the force of gravity and air resistance.

(i) Calculate the work done by the cyclist against air resistance.
mass of cyclist and bicycle = 90 kg

(4)

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Work done against air resistance = ...

(ii) State one assumption that must be made when calculating the work done against air resistance.

(1)

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(c) At the top of the slope, the road becomes horizontal. The cyclist continues to pedal with the same force.

Explain how the speed of the cyclist changes as the road becomes horizontal.

(2)

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(Total for question = 9 marks)

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Q2.

A toy car is released from rest and rolls down a slope, as shown.



mass of car = 0.160 kg
speed of car at bottom of slope = 2.6 m s^{-1}

(a) Calculate the increase in kinetic energy of the car as it accelerates down the slope.

(2)

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Increase in kinetic energy = ...

(b) As the car accelerates down the slope, the work done against frictional forces is 0.26 J.

Calculate the vertical displacement of the car.

(2)

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Vertical displacement of car = ...

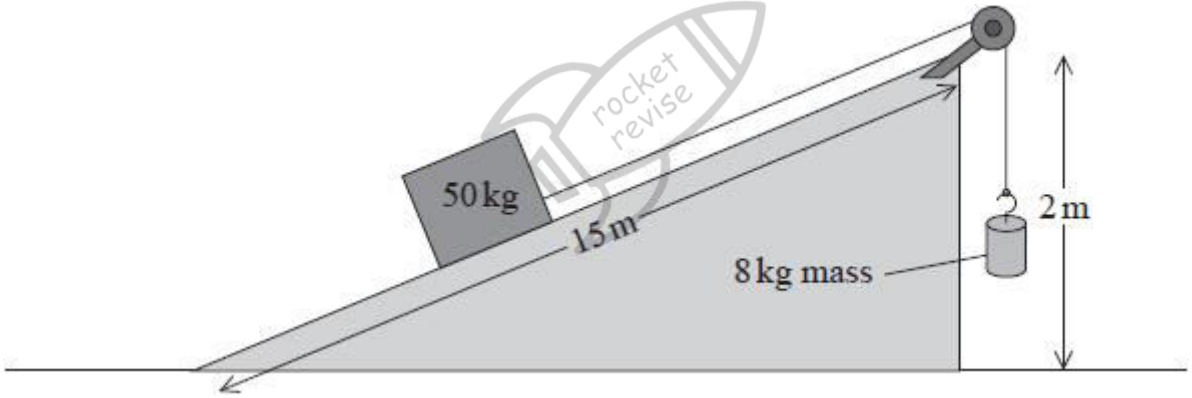
(Total for question = 4 marks)

Q3.

Machines make work easier by changing the size or direction of a force. A student designed a simple machine to lift a box of mass 50 kg. The student claimed the efficiency of the machine was greater than 90%.

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The machine used a slope of height 2.0 m and length 15 m to move the box. The box was connected to an 8.0 kg mass by a rope over a pulley as shown. As the 8.0 kg mass fell, the box moved up the slope at a steady speed.



Determine whether the maximum efficiency of the machine was greater than 90%.

(4)

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(Total for question = 4 marks)

Q4.

A regenerative braking system allows an electric car to use its kinetic energy to charge a battery as the car decelerates.

(a) A car travelling at 13.0 m s^{-1} decelerated to rest.

The energy transferred to the car's battery during the deceleration was 73.9 kJ.
Calculate the efficiency of the regenerative braking system.
mass of car = 1560 kg

(3)

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Efficiency = ...

(b) (i) Drag forces act on the car as it moves through the air.

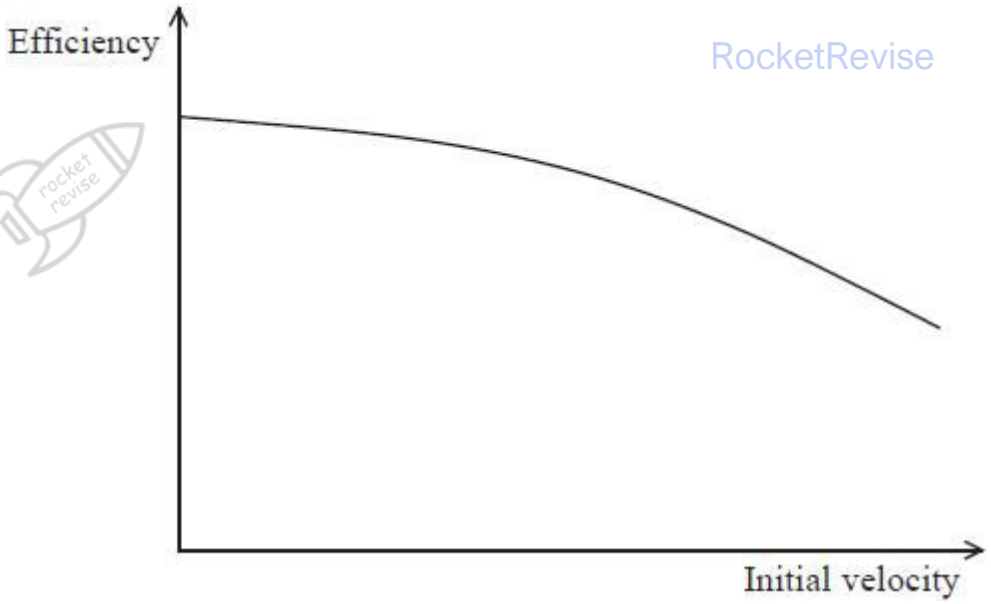
State how the drag forces vary with the velocity of the car.

(1)

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(ii) The graph shows how the efficiency of the regenerative braking system depends upon the initial velocity of the car for initial velocities in the range 20 m s^{-1} to 40 m s^{-1}.



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Explain why the efficiency of the regenerative braking system varies as shown in the graph.

(4)

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(Total for question = 8 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A lamp with an efficiency of 0.68 usefully transfers 120 J of energy.

Which of the following can be used to determine *E*, the energy supplied to the lamp?

☐ **A** $E = 0.68 \times 120 \times 100$

☐ **B** $E = 0.68 \times 120$

☐ **C** $E = \frac{120}{0.68} \times 100$

☐ **D** $E = \frac{120}{0.68}$

(Total for question = 1 mark)

Q6.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A crane lifts a container of weight 4.0×10^5N through a height of 25 m.

Which of the following gives the gravitational potential energy gained by the container in joules?

- ☐ **A** $4.0 \times 10^5 \times 9.81 \times 25$
- ☐ **B** $4.0 \times 10^5 \times 25$
- ☐ **C** $4.0 \times 10^5 \times 25 \div 9.81$
- ☐ **D** $4.0 \times 10^5 \div 9.81$

(Total for question = 1 mark)

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Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An electric pump has an output power of 25 kW and an efficiency of 52%.

Which of the following expressions gives the rate in kW at which energy is dissipated?

- ☐ **A** 0.48×25
- ☐ **B** 0.52×25
- ☐ **C** $\frac{0.48 \times 25}{0.52}$
- ☐ **D** $\frac{25}{0.52}$

(Total for question = 1 mark)

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Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A proton and an electron have equal kinetic energies. The speed of the proton is *v*.

The mass of a proton is 2000 times greater than the mass of an electron.

Which of the following expressions gives the speed of the electron?

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☐ **A** $\sqrt{2000} \times v$

☐ **B** $\frac{v}{\sqrt{2000}}$

☐ **C** $\frac{v}{2000}$

☐ **D** $2000 \times v$

(Total for question = 1 mark)

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Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A tractor pulls a trailer a distance *s* in time *t*. The useful power output of the tractor is *P*.

Which of the following equations gives the force *F* of the tractor on the trailer?

☐ **A** $F = Pts$

☐ **B** $F = \frac{Pt}{s}$

☐ **C** $F = \frac{Ps}{t}$

☐ **D** $F = \frac{st}{P}$

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A motor lifts a mass of 50 000 kg through a vertical height of 25 m.

The motor has an output power of 700 kW.

Which of the following gives the time in seconds taken to lift the mass?

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☒ **A** $\frac{50000 \times 25}{700000}$

☒ **B** $\frac{50000 \times 9.81 \times 25}{700000}$

☒ **C** $\frac{50000 \times 25}{700}$

☒ **D** $\frac{50000 \times 9.81 \times 25}{700}$

(Total for question = 1 mark)

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Q11.

Answer the question with a cross in the box you think is correct (☒). If you change your mind, put a line through the box (☒) and then mark your new answer with a cross (☒).

A machine has a power of 750 W and transfers 150 kJ of energy during a time *t*.

Which of the following expressions gives *t* in seconds?

☒ **A** $\frac{750}{150}$

☒ **B** $\frac{750 \times 150}{1000}$

☒ **C** $\frac{150 \times 1000}{750}$

☒ **D** $\frac{750}{150 \times 1000}$

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(Total for question = 1 mark)

Q12.

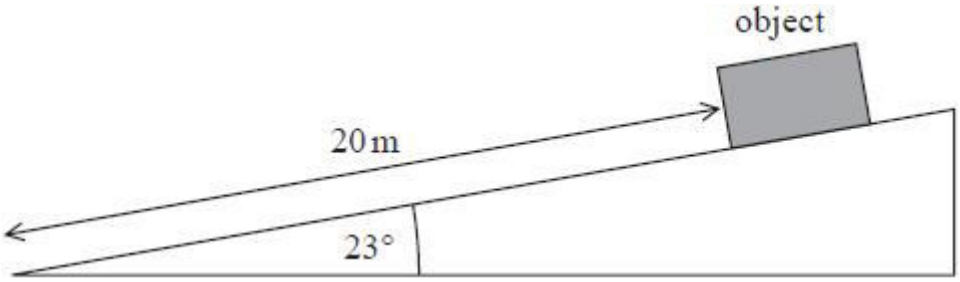
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An object accelerates from rest down a frictionless slope. The object moves a distance of 20 m. The slope

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is at an angle of 23° to the horizontal, as shown.



Which of the following gives the final speed of the object in m s^{-1}?

- ☐ **A** $\sqrt{2 \times 9.81 \times 20 \times \cos 23°}$
- ☐ **B** $\sqrt{0.5 \times 9.81 \times 20 \times \cos 23°}$
- ☐ **C** $\sqrt{2 \times 9.81 \times 20 \times \sin 23°}$
- ☐ **D** $\sqrt{0.5 \times 9.81 \times 20 \times \sin 23°}$

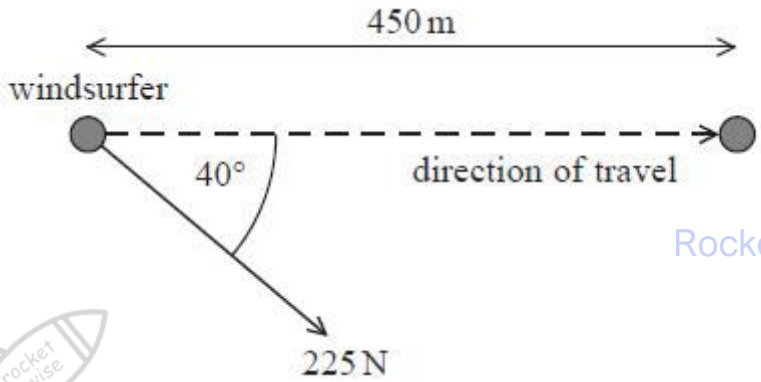
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(Total for question = 1 mark)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The wind exerts a force of 225 N on a windsurfer. The windsurfer moves a distance of 450 m at an angle of 40° to the wind, as shown.



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Which of the following expressions gives the work done, in joules, on the windsurfer?

- ☐ **A** 225 × 450
- ☐ **B** 225 × 450 × cos 40°
- ☐ **C** 225 × 450 × sin 40°
- ☐ **D** 225 × 450 × tan 40°

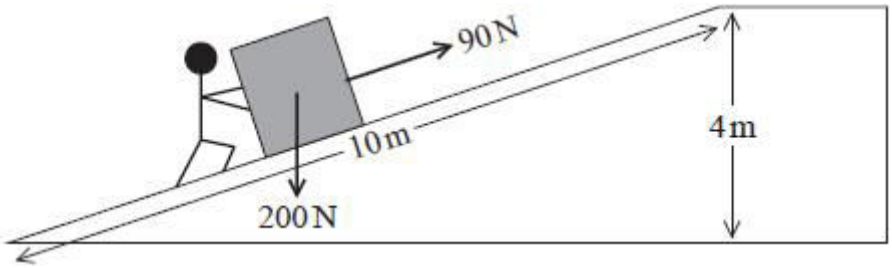
(Total for question = 1 mark)

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Q14.

A force of 90 N is used to push a box along a ramp of length 10 m, up to a platform. The platform is 4 m above the ground. The weight of the box is 200 N.



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Which of the following expressions could be used to determine the efficiency of the ramp?

☒ **A** $\frac{90 \times 10}{200 \times 4}$

☒ **B** $\frac{200 \times 4}{(90 \times 10) + (200 \times 4)}$

☒ **C** $\frac{200 \times 4}{90 \times 10}$

☒ **D** $\frac{90 \times 10}{(90 \times 10) + (200 \times 4)}$

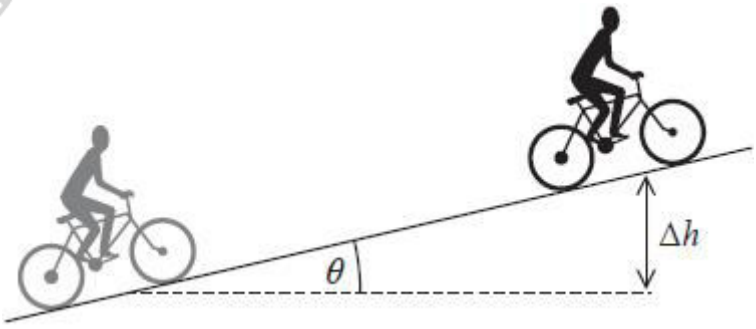
(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct (☒). If you change your mind, put a line through the box (☒) and then mark your new answer with a cross (☒).

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A cyclist rode along a slope at a constant speed. The increase in vertical height of the cyclist was Δh. The slope was at an angle θ to the horizontal as shown.



The forward force of the road on the bicycle was F and air resistance was negligible.

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Which of the following gives the work done by the cyclist?

☐ **A** $F \Delta h \cos\theta$

☐ **B** $\dfrac{F \Delta h}{\cos\theta}$

☐ **C** $F \Delta h \sin\theta$

☐ **D** $\dfrac{F \Delta h}{\sin\theta}$

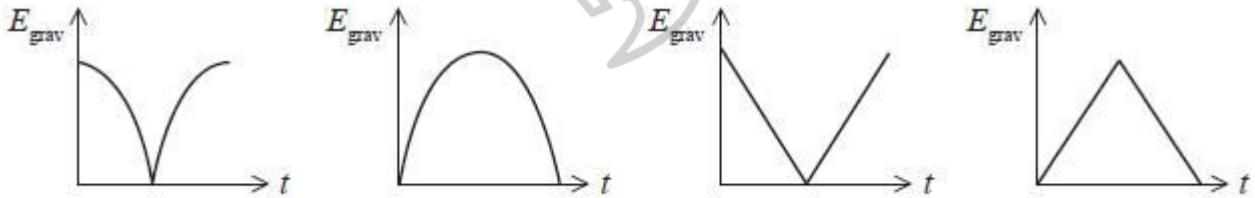
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(Total for question = 1 mark)

Q16.

A ball is dropped, bounces once and is then caught.

Which of the following graphs of gravitational potential energy E_{grav} against time t could represent the motion of the ball?



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(Total for question = 1 mark)

Q17.

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A locomotive pulls a train at constant speed against a force of 8 400 kN. The output power of the locomotive is 70 MW.

Which of the following gives the time in seconds for the locomotive to pull the train a distance of 1 km?

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☐ **A** $\frac{70 \times 10^6}{8400 \times 10^3}$

☐ **B** $\frac{70}{8400 \times 10^3}$

☐ **C** $\frac{8.4 \times 10^6 \times 10^3}{7 \times 10^7}$

☐ **D** $\frac{8400}{70 \times 1000}$

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(Total for question = 1 mark)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A water pump causes 200 g of water to be ejected from the nozzle of a garden hose each second at a velocity of 3 m s^{-1}.

Which of the following expressions could be used to determine the minimum output power in watts required from the pump?

☐ **A** $\frac{200 \times 3^2}{2}$

☐ **B** $\frac{0.2 \times 3^2}{2}$

☐ **C** $\frac{200 \times 3}{2}$

☐ **D** $\frac{0.2 \times 3}{2}$

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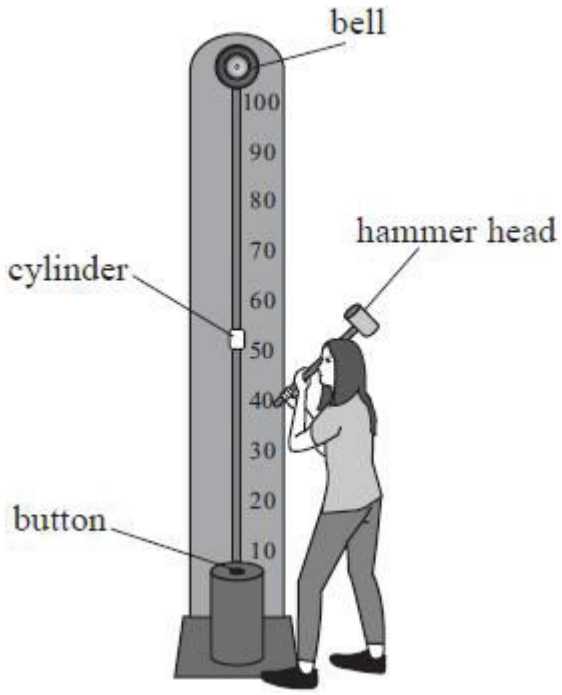
(Total for question = 1 mark)

Q19.

In a game called 'High Striker' a person hits a button with a hammer. This causes a cylinder to move towards the bell at the top, as shown.

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(a) The kinetic energy of the hammer head as it hits the button is greater than the change in gravitational potential energy of the hammer head as it moves down.

Explain why.

(3)

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(b) The hammer is initially at rest at its highest point. As the person moves the hammer head towards the button, the average resultant force on the hammer head is 58 N.

The hammer head moves a distance of 1.2 m before hitting the button. The cylinder must move 2.7 m upwards to hit the bell.
Deduce whether the cylinder hits the bell.
mass of cylinder = 0.15 kg
efficiency of energy transfer = 4.0 %

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(5)

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(c) If the velocity of the hammer head as it hits the button doubles, the height gained by the cylinder does **not** double.

Explain why.

(2)

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(Total for question = 10 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An object moves with a velocity v.

The kinetic energy of the object is doubled.

Which of the following is an expression for the new velocity of the object?

☐ **A** $\sqrt{2}v$

☐ **B** $2v$

☐ **C** $2\sqrt{2}v$

☐ **D** $4v$

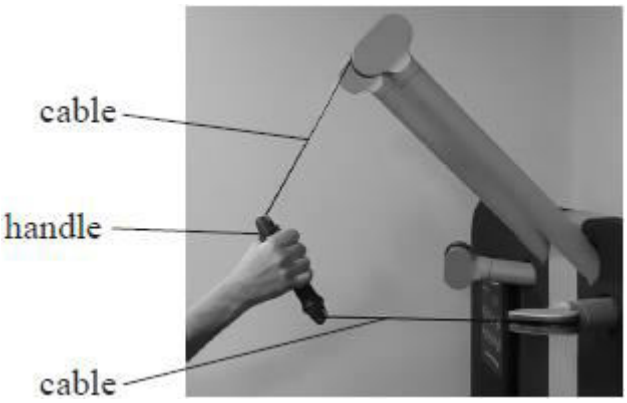
(Total for question = 1 mark)

Q21.

A student exercises by lifting weights attached to a machine. The machine has a cable, as shown in the photograph. The cable passes through a handle and is connected to a weight.

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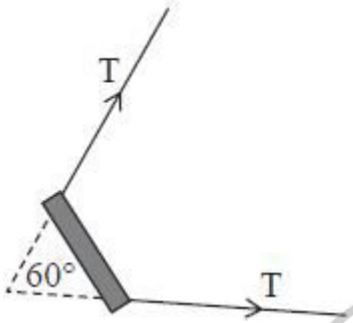
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There is a tension *T* in the cable. The angle between the two sides of the cable is 60° as shown in the diagram.

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Not to scale



(a) A person keeps the cable stationary by applying a force *F* of 121 N.

Determine the magnitude of the tension *T* in the cable using a scaled vector diagram.

(4)

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T = ...

(b) The cable is attached to a 150 N weight. Each time the person pulls the handle the weight moves up a vertical distance of 0.25 m. The weight returns to its original position when the handle is released.

The person exercises by pulling and releasing the handle. They do this 90 times in 120 s. The exercise machine displays the average power of the person as 35 W.
Deduce whether the power displayed by the machine is consistent with this data.

(3)

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(Total for question = 7 marks)

Q22.

Every litre of fuel used in a car engine releases 32 MJ of energy. During one journey the car used 65 litres of fuel. The car journey lasted 8 hours.

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(a) Show that the average power input to the engine was about 70 kW.

(2)

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(b) The car travelled a total distance of 730 km during the journey.

The average force of the engine was 2.1 kN.
Show that the average power output of the engine was about 50 kW.

(3)

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(c) Determine the efficiency of the engine.

(2)

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Efficiency of engine = ...

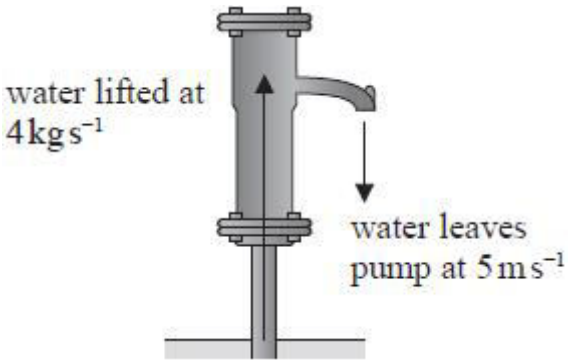
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(Total for question = 7 marks)

Q23.

An electric pump lifts water from a well through a height of 10 m at a rate of 4 kg s^{-1}.
The water leaves the pump with a velocity of 5 m s^{-1} as shown.



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Which of the following expressions could be used to determine the minimum power output of the pump in watts?

☐ **A** $\frac{1}{2}(4 \times 5^2)$

☐ **B** $(4 \times 9.81 \times 10)$

☐ **C** $(4 \times 9.81 \times 10) + \frac{1}{2}(4 \times 5^2)$

☐ **D** $(4 \times 9.81 \times 10) - \frac{1}{2}(4 \times 5^2)$

(Total for question = 1 mark)

Q24.

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The engine of a motor boat provides a constant horizontal force. As the velocity of the boat increases, the resultant horizontal force on the boat decreases. Eventually the boat will travel at its terminal velocity.

The constant horizontal force from the engine is 5.5 kN.

(a) State what is meant by a resultant force.

(1)

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(b) At a certain time the drag force on the boat is 3.1 kN.

Calculate the acceleration of the boat at this time.

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mass of boat = 7.5×10^3 kg

(2)

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Acceleration = ...

(c) The terminal velocity of the boat is 4.8 m s^{-1}.

Calculate the output power of the engine.

(2)

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Output power = ...

(Total for question = 5 marks)

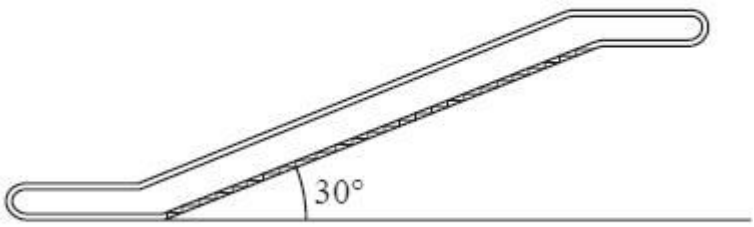
Q25.

Moving walkways are often found in airports. One moving walkway carries passengers up an incline of 30°, as shown.



(Source: © ilolab/Shutterstock.)

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(a) A single passenger of mass 72 kg stands on the walkway.
The speed of the walkway is 0.51 m s^{-1}.

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Show that the rate at which the walkway does work on the passenger is about 200 W.

(3)

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(b) The walkway system has an efficiency of 78%.

Calculate the power input to the system when 15 passengers of average mass 72 kg are standing on the walkway.

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(3)

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Power input = ...

(Total for question = 6 marks)

Q26.

A high-speed train accelerates from rest along a straight horizontal track.

(a) The final speed of the train is 76 m s^{-1}.

Show that the useful work done to accelerate the train is about 2 GJ.
mass of train = 7.2×10^5 kg

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(2)

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(b) The train accelerates for 180 s. The train has an input power of 16 MW while accelerating.

Determine the work done against air resistance as the train accelerates.

(2)

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Work done against air resistance = ...

(Total for question = 4 marks)

Q27.

Which of the following quantities has the SI base units kg m^2 s^{-3}?

☒ **A** force

☒ **B** momentum

☒ **C** power

☒ **D** work done

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(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A power station provides an electrical output power of 2.1 GW. The power station transfers thermal energy to the surroundings at a rate of 3.4 GW.

Which of the following gives the efficiency of the power station?

☒ **A** $\frac{2.1}{3.4 + 2.1}$

☒ **B** $\frac{3.4}{3.4 + 2.1}$

☒ **C** $\frac{3.4 - 2.1}{2.1}$

☒ **D** $\frac{3.4 - 2.1}{3.4}$

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(Total for question = 1 mark)

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Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A hydroelectric power station has an efficiency of 32%. In one hour the useful energy output of the power station
is 1.2×10^{13} J.

Which of the following expressions gives the total power input to the power station in watts?

☐ **A** $1.2 \times 10^{13} \times 0.32$

☐ **B** $\dfrac{1.2 \times 10^{13}}{0.32 \times 3600}$

☐ **C** $\dfrac{1.2 \times 10^{13} \times 0.32}{3600}$

☐ **D** $\dfrac{1.2 \times 10^{13}}{0.32}$

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(Total for question = 1 mark)

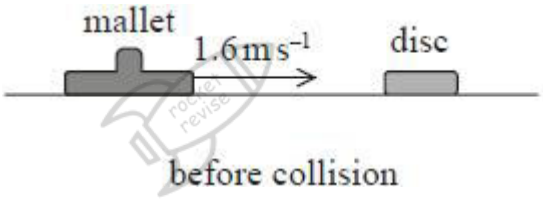
Q30.

A game is played where each player must use a 'mallet' to hit a disc across a table into the opponent's goal.

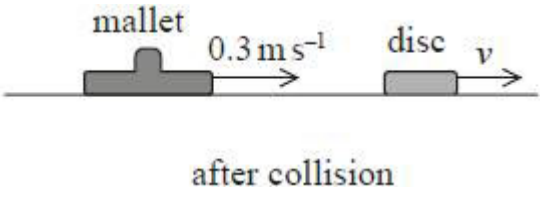
One player accidentally lets go of a mallet. The mallet travels at a speed of 1.6 m s^{-1} and collides with a stationary disc.

After the collision, the mallet continues in the same direction at a lower speed of 0.3 m s^{-1}. The disc moves in the same direction as the mallet with a velocity *v*, as shown.

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before collision



after collision

(a) Calculate a value for *v*.

Assume that frictional forces are negligible.
mass of disc = 0.035 kg
mass of mallet = 0.17 kg

(3)

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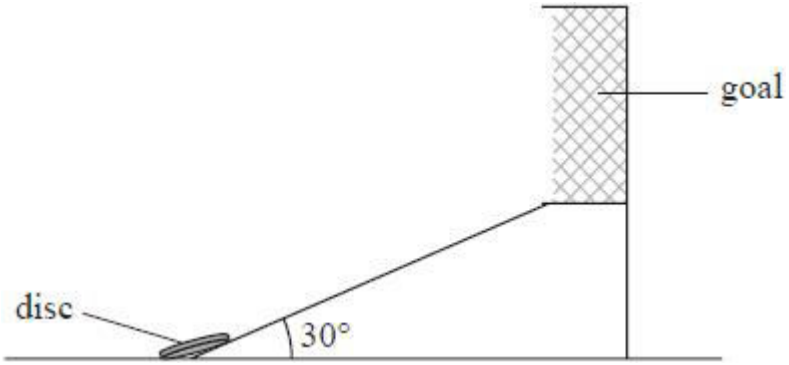
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v = ...

(b) To reach the goal, the disc must move up a ramp which is at an angle of 30° to the horizontal.



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The velocity of the disc at the bottom of the ramp is 5.0 m s^{-1}. The disc moves up the ramp and work is done by the disc against the frictional force. The disc moves a distance of 6.5 cm up the ramp before moving back down.
Determine the frictional force acting on the disc.

(6)

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Frictional force = ...

(Total for question = 9 marks)

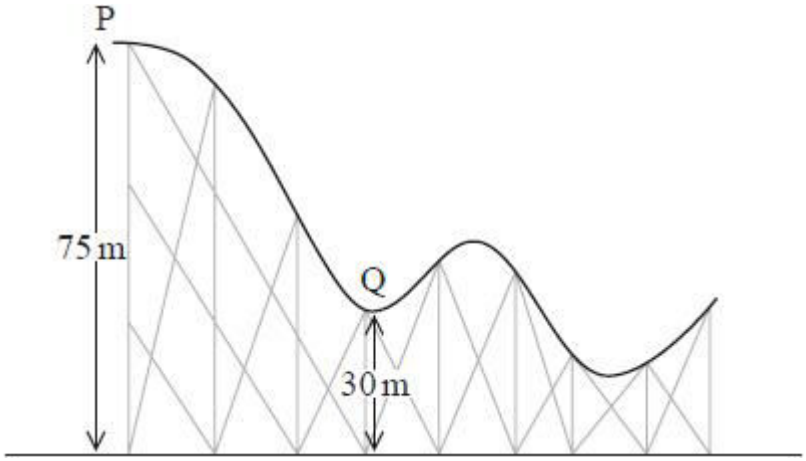
Q31.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The diagram shows a roller coaster. A roller coaster car stops momentarily at P before descending towards Q.



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Which of the following expressions could be used to determine the velocity of the roller coaster car at Q?

☐ **A** $\sqrt{75g} - \sqrt{30g}$

☐ **B** $\sqrt{150g} - \sqrt{60g}$

☐ **C** $\sqrt{45g}$

☐ **D** $\sqrt{90g}$

(Total for question = 1 mark)

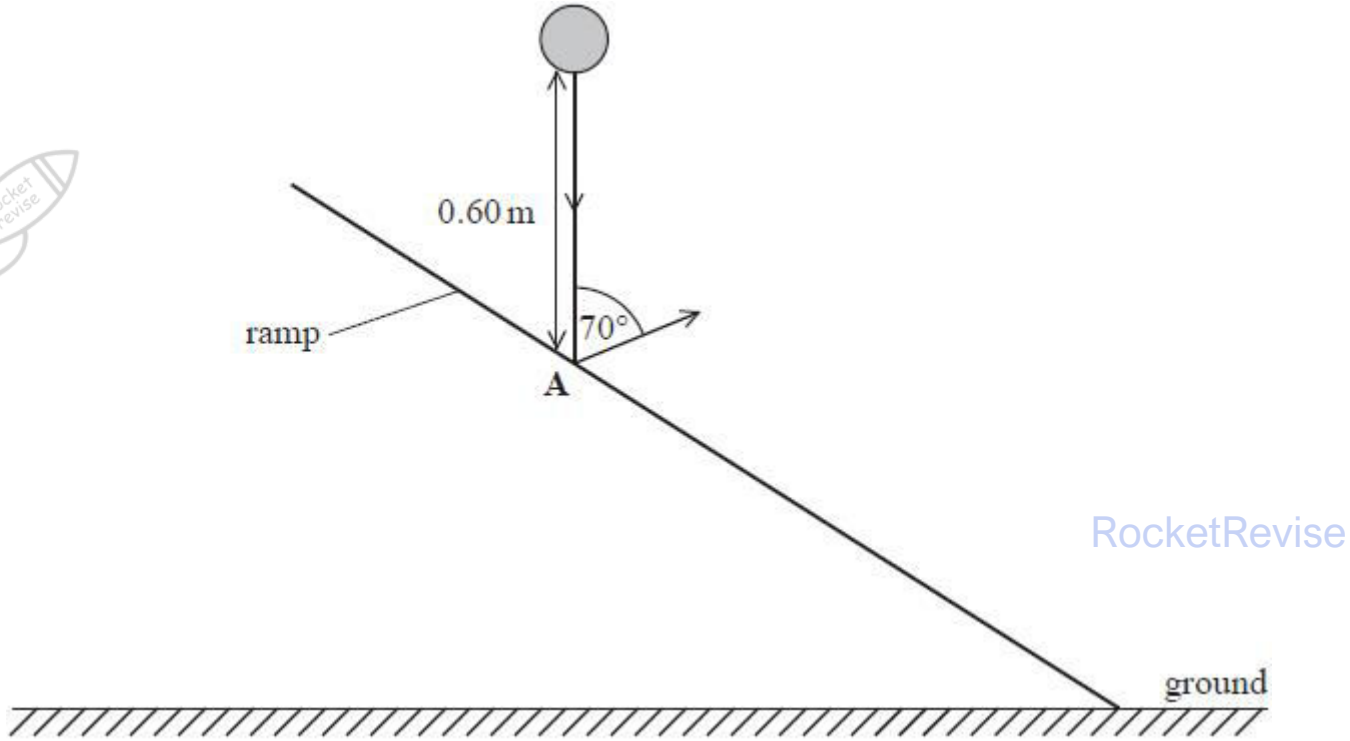
Q32.

A ball falls through a vertical height of 0.60 m before bouncing at point A on a ramp, as shown.

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(a) Show that the velocity of the ball immediately before the bounce is about 3 m s^{-1}.

(2)

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(b) Kinetic energy is conserved as the ball bounces off the ramp. The ball bounces at an angle of 70° to the vertical.

State expressions for the horizontal and vertical components of velocity of the ball immediately after the bounce.

(2)

Horizontal component = ...

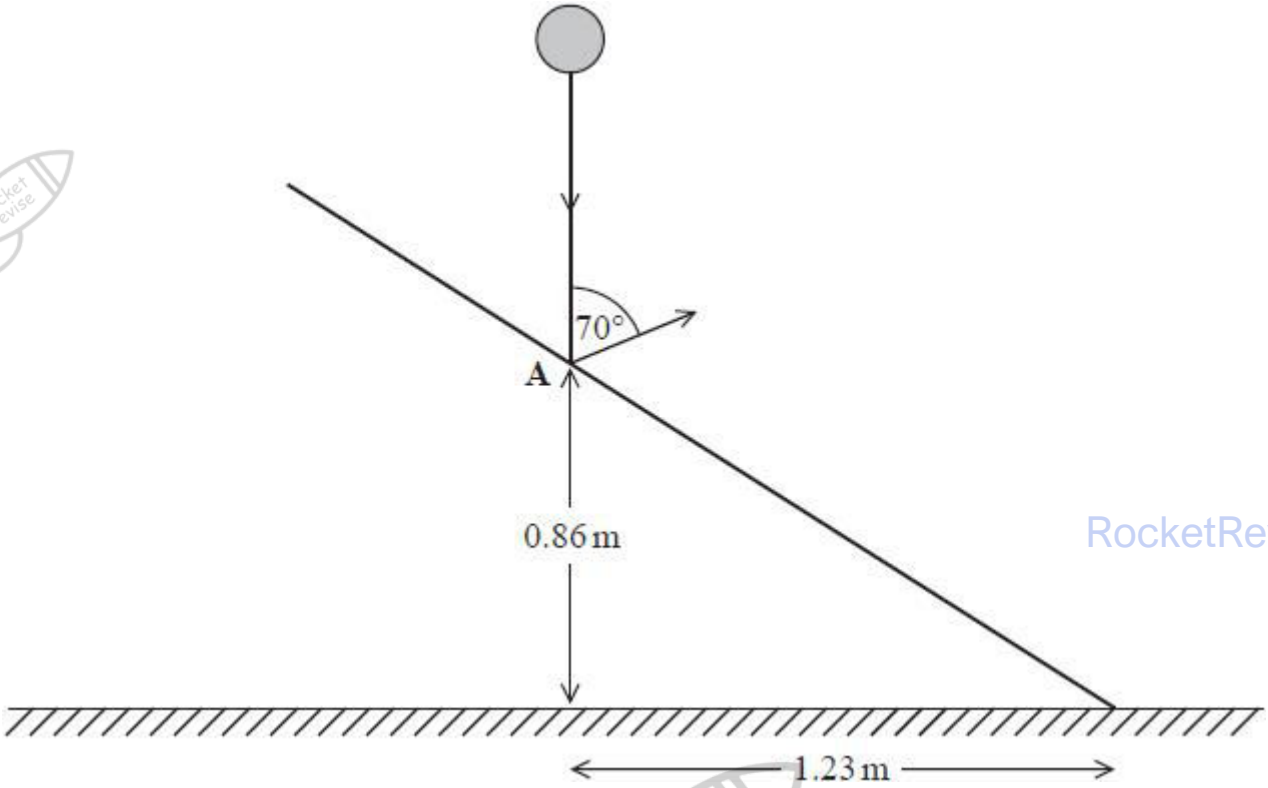
Vertical component = ...

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(c) Point A is 0.86 m vertically above the ground and 1.23 m horizontally from the end of the ramp as shown.

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Deduce whether the ball will bounce a second time on the ramp.

(4)

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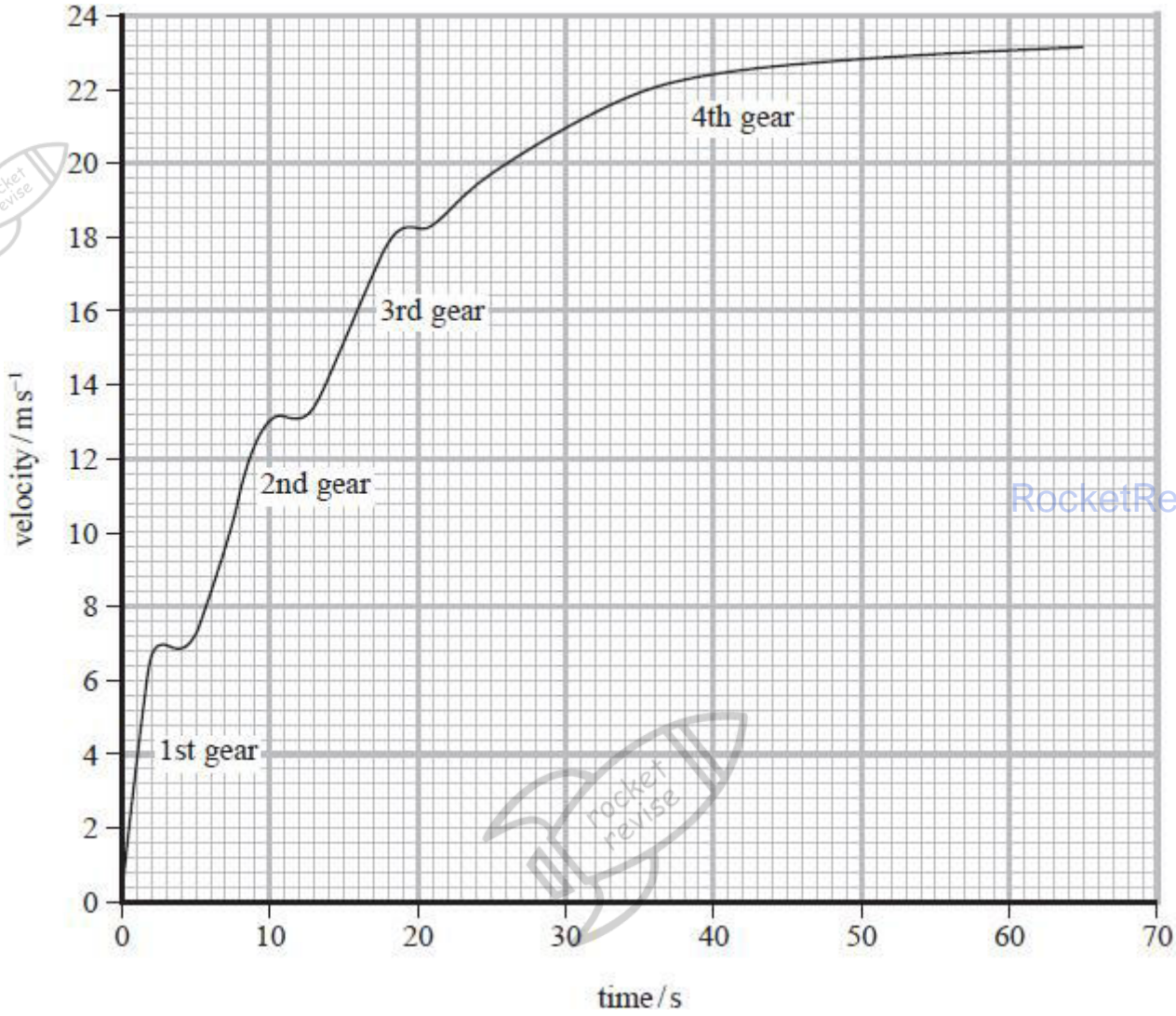
(Total for question = 8 marks)

Q33.

The velocity-time graph for a car for the first part of a journey is shown.

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(a) As the velocity of the car increased, the driver changed gear. Each change in gear produced a small decrease in velocity.

(i) State how the graph shows that in 3rd gear the car has a smaller acceleration than in 1st gear.

(1)

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(ii) Determine the average acceleration when the car is accelerating in 2nd gear.

(2)

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Average acceleration in 2nd gear = ...

(iii) The equation that links the power P of a car engine to the forward force F on the car and the velocity v of the car is

$$P = Fv$$

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Comment on the magnitude of *F* when the car is in higher gears, assuming *P* remains constant.

(2)

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(b) The manufacturer of a different car gives the following information.

Gear	Maximum velocity / m s^{-1}	Maximum acceleration / m s^{-2}
1st	18	2.9
2nd	32	1.2
3rd	46	0.83
4th	74	0.72

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(i) Calculate the minimum time taken for the car to accelerate from rest to 60 miles per hour. You may assume that the time taken to change gears is negligible.

1 mile = 1600 m

(4)

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Time = ...

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(ii) Explain why there will be a maximum speed at which the car can travel.

(3)

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(Total for question = 12 marks)

Q34.

A motor with an efficiency of 68% is used to raise a load of 350 N through a height of 25 m.

Which of the following expressions gives the energy supplied to the motor?

☐ **A** $\frac{68}{350 \times 25}$

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☐ **B** $\frac{0.68}{350 \times 25}$

☐ **C** $\frac{350 \times 25}{68}$

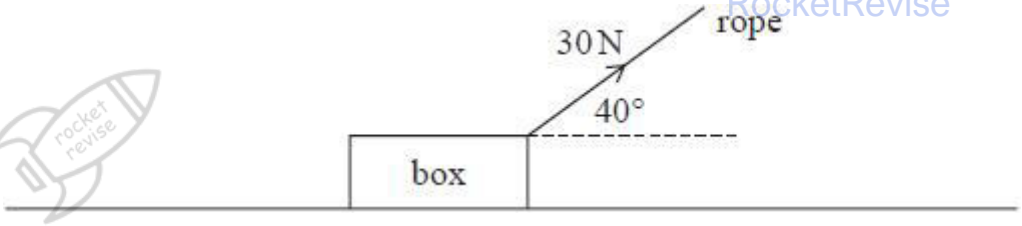
☐ **D** $\frac{350 \times 25}{0.68}$

(Total for question = 1 mark)

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A box is pulled along the ground by a rope inclined at an angle of 40° to the horizontal as shown.



The tension in the rope is 30 N.

The box is pulled 5 m along the ground.

Which of the following is the work done on the box?

☐ **A** 150 J
☐ **B** 115 J
☐ **C** 96 J

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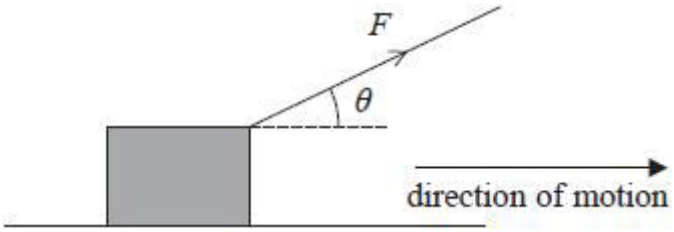
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☒ **D** 23 J

(Total for question = 1 mark)

Q36.

A rope is used to pull a box a fixed distance s along a horizontal surface. The rope is at an angle θ to the horizontal and a constant force F is applied to the rope as shown.



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Explain how the work done on the box by F varies as θ varies.

(3)

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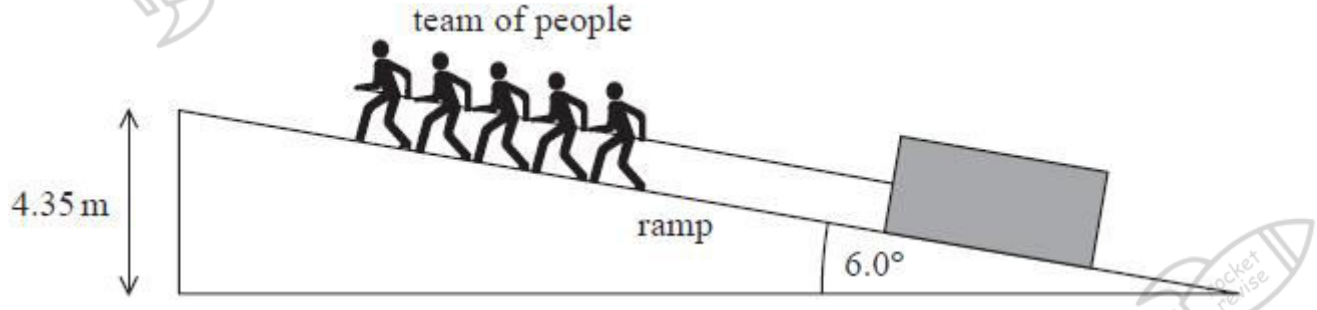
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(Total for question = 3 marks)

Q37.

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The diagram shows a system used to move a stone block up a ramp. A team of people uses a rope to pull the block at a constant speed.



height of ramp = 4.35 m

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angle of ramp to horizontal = 6.0°
mass of block = 2.10×10^3 kg
speed of block up ramp = 0.450 m s^{-1}
total power of team = 6.25 kW

(a) Show that the total force that the team exerts on the block is about 14 kN.

(2)

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(b) Determine the total work done by the team.

RocketRevise

(3)

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Total work done = ...

(c) Show that the useful work done on the block is about 90 kJ.

(2)

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(d) Determine the efficiency of the system.

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(2)

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Efficiency = ...

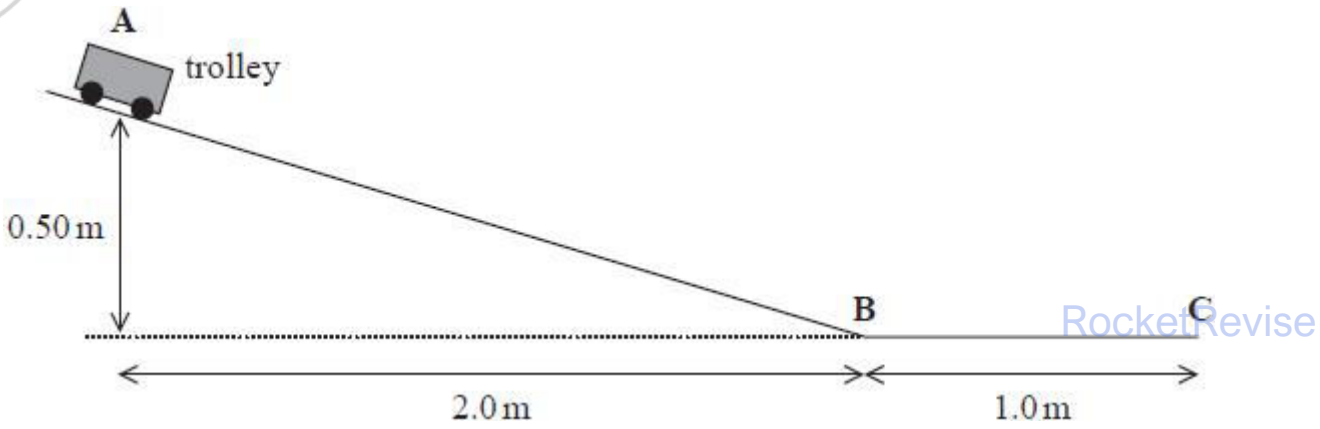
(Total for question = 9 marks)

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Q38.

A trolley accelerates from rest at point A, down a straight track to point B. The trolley then continues along a horizontal track to point C, as shown.



The effects of air resistance and friction are negligible.

(a) Show that the trolley reaches point B with a speed of about 3 m s^{-1} .

(3)

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(b) Determine the time taken by the object to move from point A to point C.

(4)

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Time taken = ...

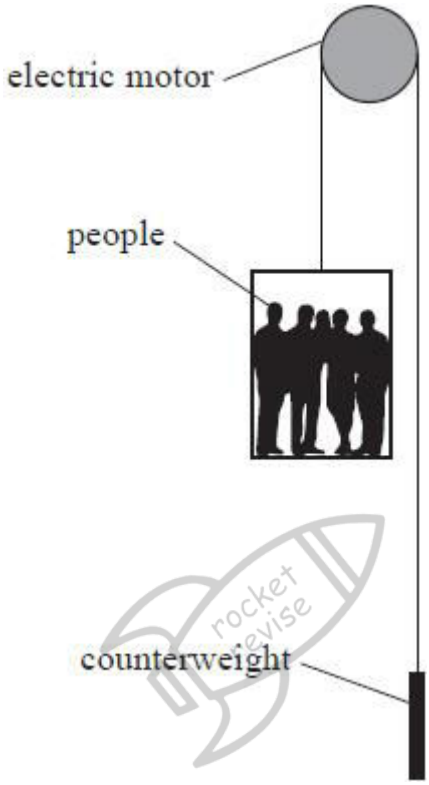
(Total for question = 7 marks)

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Q39.

The diagram shows a lift system for moving people up and down a tall building. There is a counterweight to balance the weight of the lift. An electric motor is used to raise and lower the lift.



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(a) Explain how the counterweight affects the amount of work required from the electric motor to raise the lift.

(2)

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(b) The electric motor raises the lift through a height of 40.0 m in a time of 30.0 s.

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Show that the output power of the electric motor is about 12 kW.
total mass of lift and people = 2 250 kg
mass of counterweight = 1 300 kg

(4)

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(c) The electric motor dissipates energy to the surroundings at a rate of 3 600 W.

Determine the efficiency of the electric motor.

(2)

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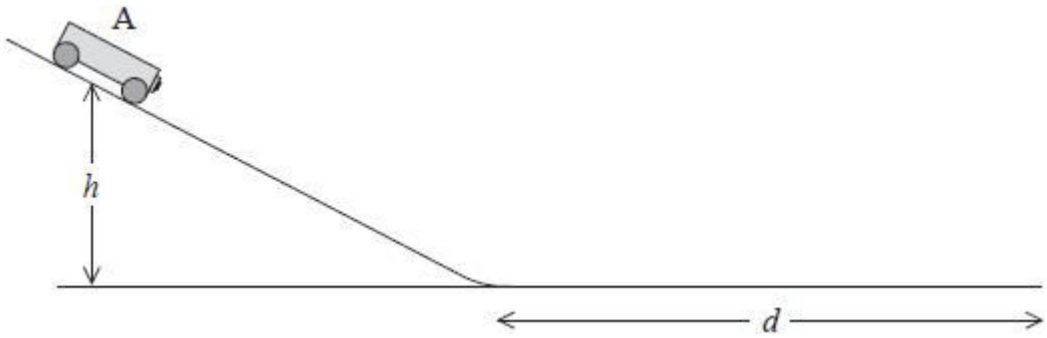
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Efficiency = ...

(Total for question = 8 marks)

Q40.

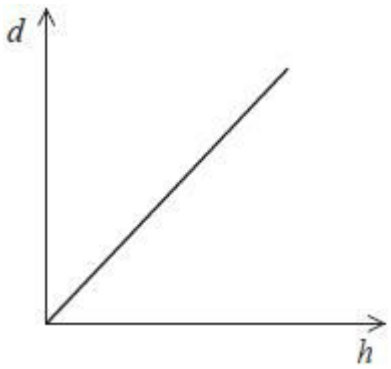
A student set up the apparatus shown.



(a) When released from a height *h*, trolley A ran down the slope and then continued to move horizontally. On the horizontal part of the track a frictional force *F* brought the trolley to rest over a short distance *d*. The trolley has a mass *m*. The student measured *d* for a range of heights *h*.

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The student plotted the following graph of *d* against *h*.



Derive an expression for the gradient of the graph, in terms of *F*, *m* and *g*.

(2)

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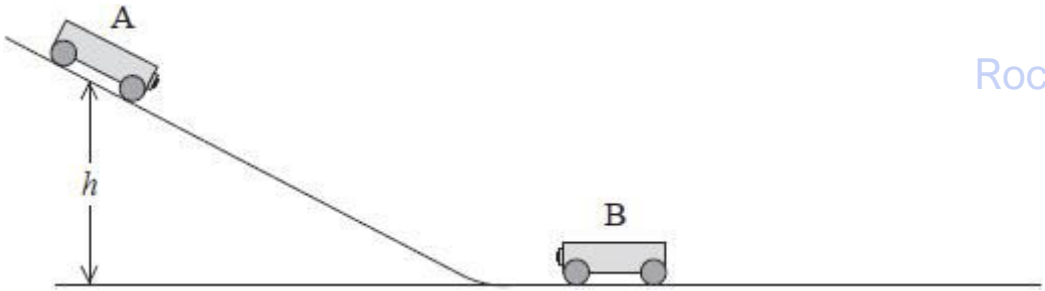
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(b) In a second experiment, an identical trolley B was placed at rest at the bottom of the slope. When trolley A was released as before, it rolled down and collided with trolley B. After the collision the two trolleys joined together and moved off to the right with a velocity *v*.



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The student predicted that, provided friction was ignored, $v = \sqrt{\frac{gh}{2}}$.
Assess whether the student was correct.

(4)

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(Total for question = 6 marks)

Q41.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A helicopter moves vertically upwards with an increasing speed.

Which of the following describes the increase in gravitational potential energy of the helicopter?

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- **A** The work done on the helicopter by the force of gravity.
- **B** The total work done by the helicopter as it moves upwards.
- **C** The work done by the helicopter against the force of gravity.
- **D** The work done by the helicopter to increase speed.

(Total for question = 1 mark)

Q42.

An object of mass 8.2 kg, initially at rest, falls a vertical distance of 25 m through the air and has a final velocity of
20 m s^{-1}.

Which of the following gives the energy in joules dissipated by air resistance?

- **A** $8.2 \times 9.81 \times 25$
- **B** $0.5 \times 8.2 \times 20^2 + 8.2 \times 9.81 \times 25$
- **C** $8.2 \times 9.81 \times 25 - 0.5 \times 8.2 \times 20^2$
- **D** $0.5 \times 8.2 \times 20^2 - 8.2 \times 9.81 \times 25$

(Total for question = 1 mark)

Q43.

Which of the following is equivalent to 1 kilowatt-hour?

- **A** 0.28 J
- **B** 0.28 W
- **C** 3.6×10^6J
- **D** 3.6×10^6W

(Total for question = 1 mark)

Q44.

A train moves up an inclined track at constant velocity. The train moves through a vertical height of 25 m.

(a) Show that the gravitational potential energy gained by the train is about 400 MJ.

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mass of train = 1.7×10^6 kg

(2)

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(b) The train's engine has an output power of 20 MW. The train takes a time of 35 s to move through the vertical height of 25 m.

Calculate the work done by the engine against resistive forces.

(3)

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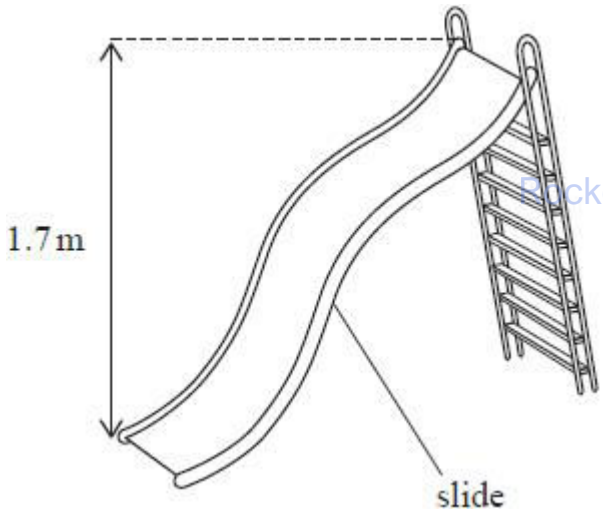
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Work done = ...

(Total for question = 5 marks)

Q45.

A child's slide has a vertical height of 1.7 m as shown.



(a) A child starts from rest at the top of the slide and moves down the slide.

The resistive forces acting on the child can be ignored.

(i) Explain why the equations of motion should **not** be used to determine the maximum speed of the child.

You should consider the shape of the slide.

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(3)

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(ii) The child has a mass of 24 kg.
Determine the maximum speed v of the child on the slide.

(3)

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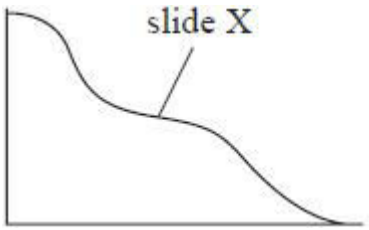
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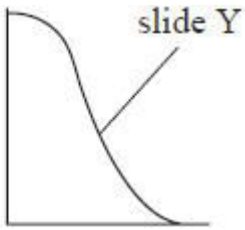
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v = ...

(b) Two slides, X and Y, have the same height.

Slide X is longer than slide Y, as shown.





A child moves from rest down slide X and reaches a maximum speed v_X.
A second child moves from rest down slide Y and reaches a maximum speed v_Y.
The child on slide Y has a greater mass than the child on slide X.
The resistive forces acting on both children can be ignored.
Explain how v_X compares with v_Y.

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(3)

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(Total for question = 9 marks)

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Q46.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

An engine is used to remove water from a mine. The engine transfers energy to the water at a rate of 180 kW and wastes energy at a rate of 60 kW.

Which of the following expressions gives the efficiency of the engine?

☐ **A** $\frac{180}{240}$

☐ **B** $\frac{60}{240}$

☐ **C** $\frac{60}{180}$

☐ **D** $\frac{180}{300}$

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(Total for question = 1 mark)

Q47.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A cyclist moves along a horizontal road at a constant speed. She then moves at the same speed up a hill with a constant gradient.

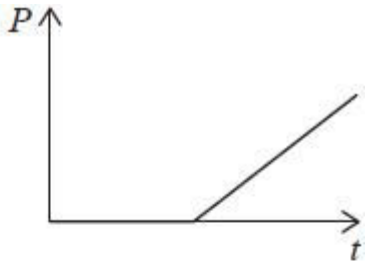
Which of the following graphs shows how the power output P of the cyclist varies with time t?

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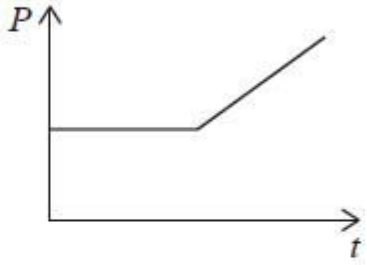
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☒ **A**

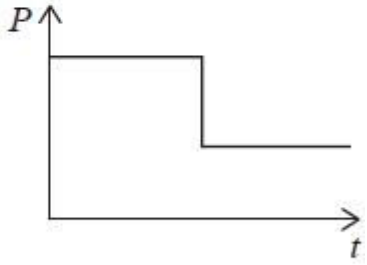


☒ **B**

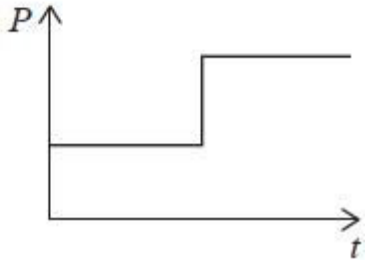


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☒ **C**



☒ **D**



(Total for question = 1 mark)

Q48.

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In a high jump competition, athletes jump over a bar.

An athlete has a weight of 680 N.

(a) The athlete exerts a force of 890 N on the ground as he jumps into the air.

Explain why the athlete accelerates upwards.
Your answer should refer to Newton's third law.

(3)

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(b) The athlete moves horizontally as he passes over the bar.

(i) The kinetic energy of the athlete is 409 J.
Show that the athlete has a horizontal speed of about 3.4 m s^{-1}.

(3)

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(ii) After passing over the bar, the athlete moves through a vertical distance of 1.4 m before he lands.
Calculate the velocity of the athlete as he lands.

(5)

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Magnitude of velocity = ...

Angle from vertical = ...

(Total for question = 11 marks)

Q49.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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Which of the following units is equivalent to the watt?

- ☒ **A** N m
- ☒ **B** N m^{-1}
- ☒ **C** J s
- ☒ **D** J s^{-1}

(Total for question = 1 mark)

Q50.

RocketRevise

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A person stands on a moving staircase.

The moving staircase increases the gravitational potential energy of the person by 5000 J in a time of 42 s.

The efficiency of the moving staircase is 0.63

Which of the following expressions gives the power input to the moving staircase in watts?

- ☒ **A** $\frac{42}{5000 \times 0.63}$
- ☒ **B** $\frac{42 \times 0.63}{5000}$
- ☒ **C** $\frac{5000}{42 \times 0.63}$
- ☒ **D** $\frac{5000 \times 0.63}{42}$

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(Total for question = 1 mark)

Q51.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The driver of a car applies the brakes. The frictional force *F* between the tyres and the road is constant.

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The average power dissipated by *F* is 50 kW.

The car moves a distance of 13 m in a time of 2.3 s before coming to rest.

Which of the following expressions gives *F* in newtons?

☐ **A** $\frac{50 \times 10^3 \times 2.3}{13}$

☐ **B** $\frac{50 \times 10^3 \times 13}{2.3}$

☐ **C** $\frac{13}{50 \times 10^3 \times 2.3}$

☐ **D** $\frac{2.3}{50 \times 10^3 \times 13}$

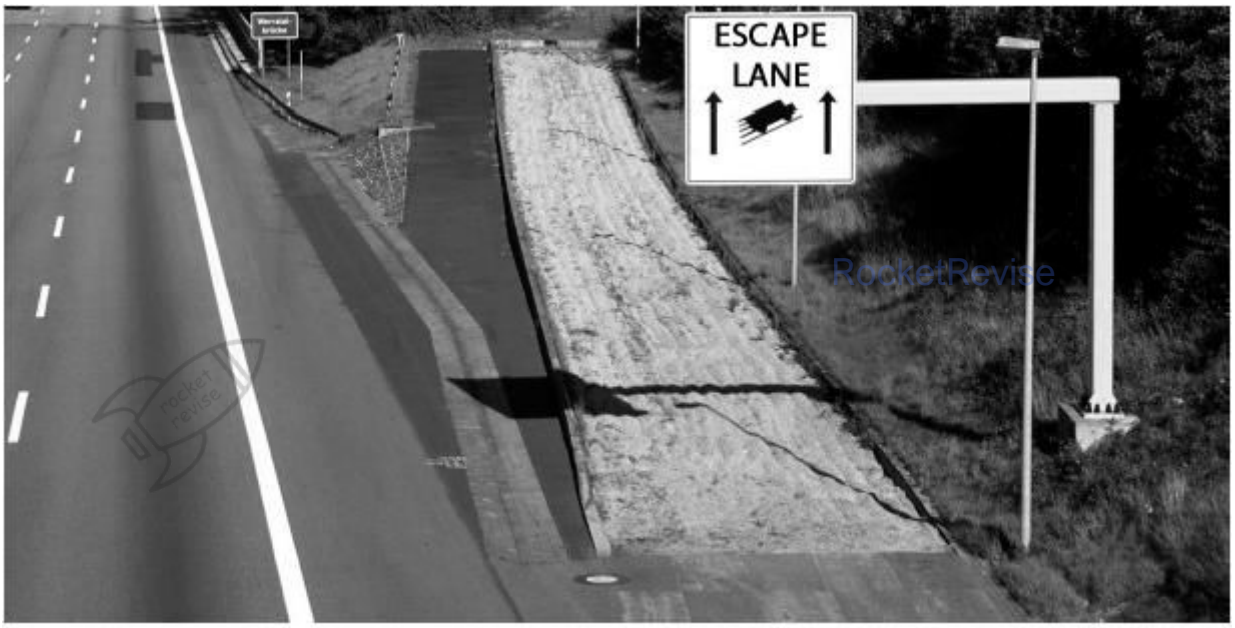
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(Total for question = 1 mark)

Q52.

An 'escape lane' can stop cars or trucks safely if the brakes stop working when moving downhill.

The photograph shows an escape lane.

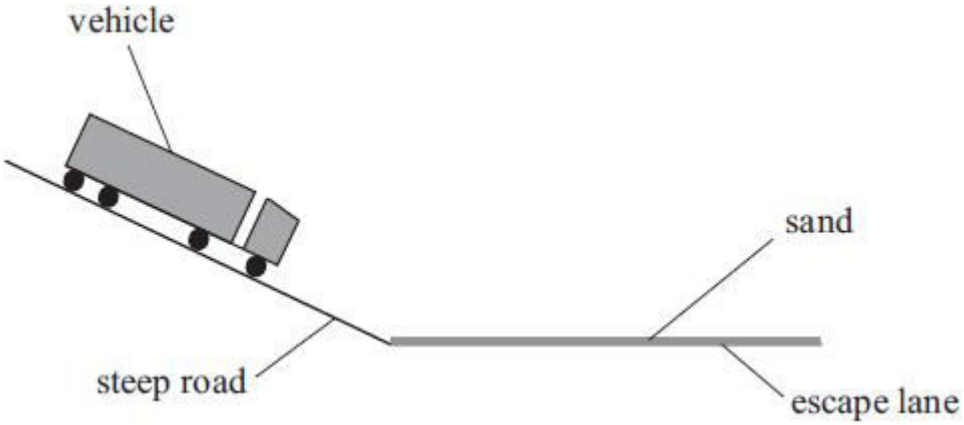


(Source: adapted from © Presse03)

The escape lane is made of sand and is horizontal, as shown below. The resistive force of the sand on a vehicle reduces the kinetic energy of the vehicle.

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(a) As a truck enters the escape lane, the kinetic energy of the truck is 8.0×10^6 J.

The sand exerts an average resistive force of 3.5×10^5 N on the truck as the truck comes to rest.
Calculate the distance moved by the truck on the escape lane.

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(2)

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Distance = ...

(b) The resistive force exerted by the sand on a vehicle is proportional to the weight of the vehicle.

The weight of a car is less than the weight of a truck.
Explain how the deceleration of a car on the escape lane would compare with the deceleration of a truck.

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(Total for question = 4 marks)

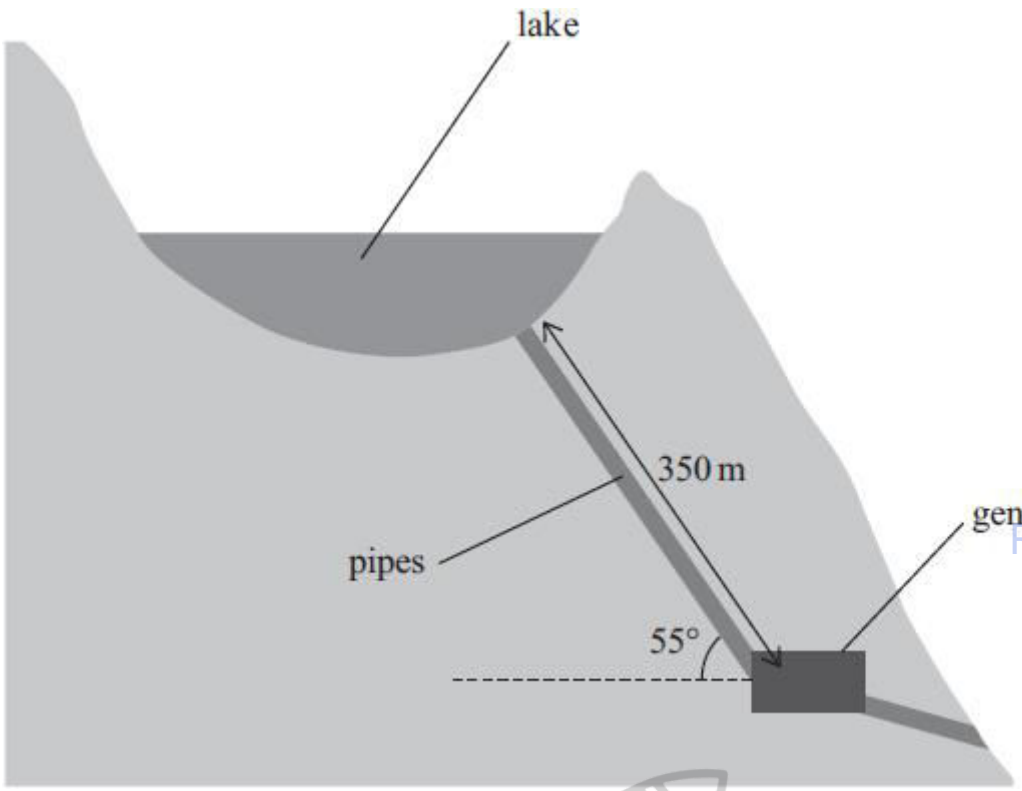
Q53.

The Cruachan hydroelectric power station in Scotland is built inside a mountain.

Water from a lake near the top of the mountain flows through straight pipes to a generator. The pipes have a length of 350 m and are inclined at an angle of 55° to the horizontal, as shown.

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The gravitational potential energy of the water in the lake is used to generate electrical power.

(a) In one minute, each pipe transfers 1700 m^3 of water to the generator. The density of the water in the lake is 1100 kg m^{-3}.

Show that the power transferred by the water in each pipe is about 88 MW.

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(b) Water travels from the lake to the generator through 4 identical pipes.

As water moves through the 4 pipes to the generator, the total power wasted is 256 MW.
Calculate the efficiency of the power station.

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Efficiency = ...

(Total for question = 7 marks)

Q54.

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A skydiver jumped from an aeroplane.

(a) The initial vertical velocity of the skydiver was zero.

During the first 1.5 s, the skydiver moved a vertical distance of 11 m.
Air resistance may be considered negligible if the mean acceleration of the skydiver is greater than 0.95 g.
Deduce whether the air resistance acting on the skydiver was negligible during the first 1.5 s.

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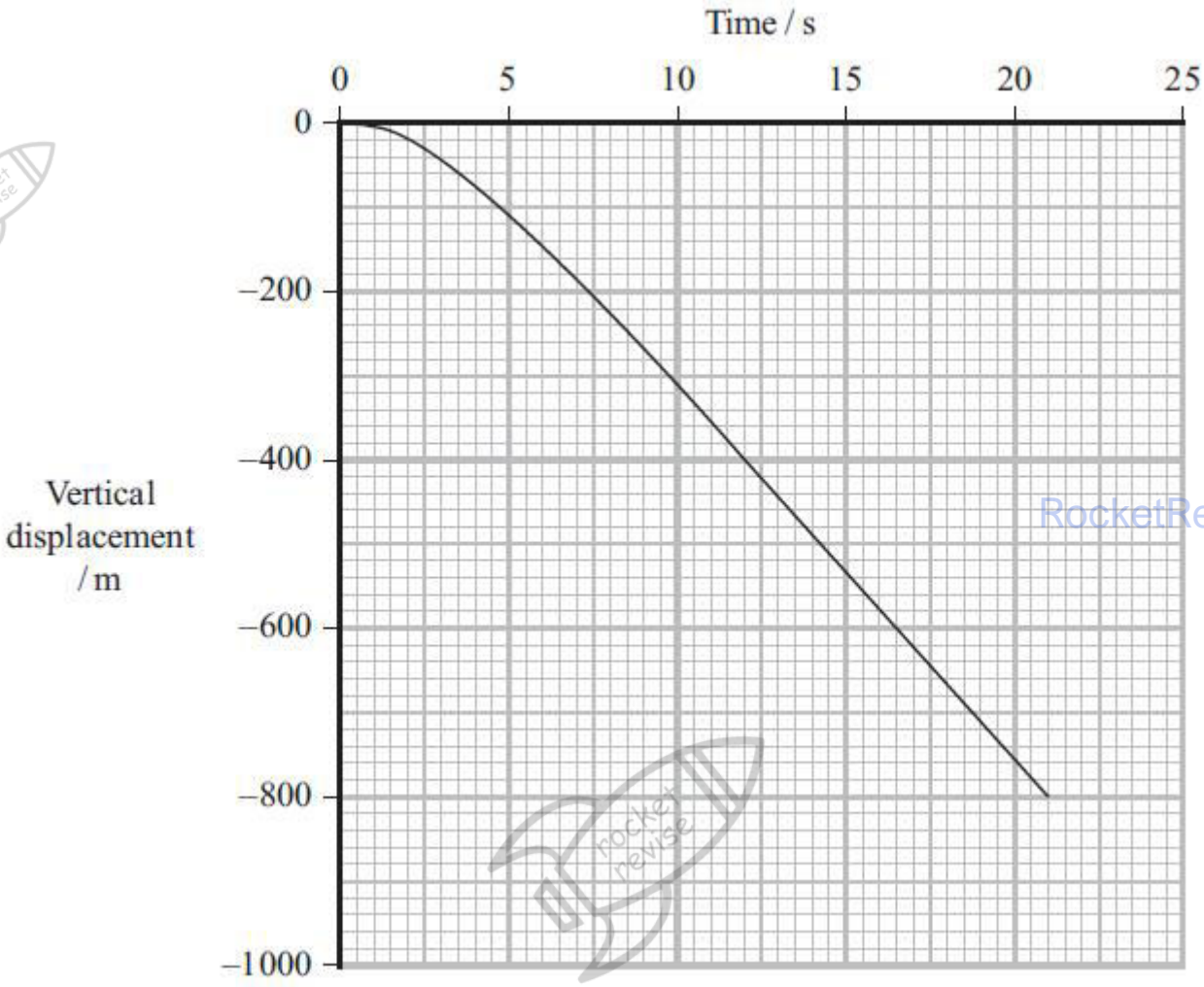
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(b) The displacement-time graph for the skydiver between jumping from the aeroplane and opening his parachute is shown.

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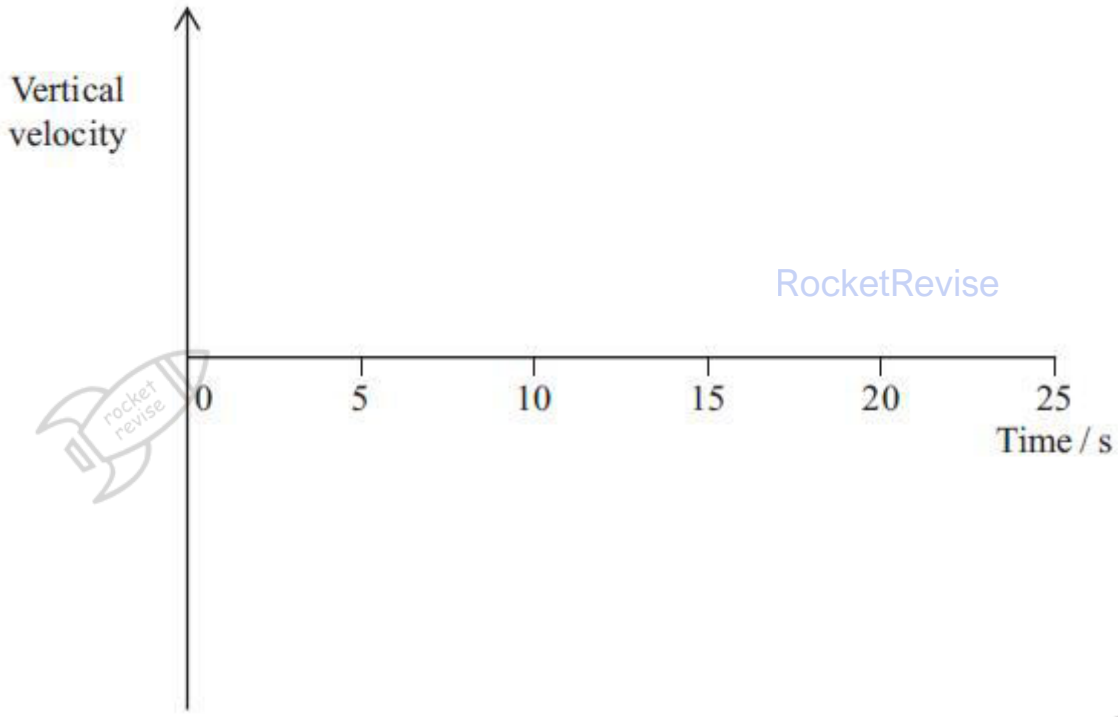
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(2)

Sketch the velocity-time graph, on the axes below, for the skydiver between jumping from the aeroplane and opening his parachute.



(c) After jumping from the aeroplane, the skydiver accelerated and then reached a terminal velocity

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before opening his parachute.

Explain how energy conservation applies to the skydiver before opening his parachute.

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(d) The skydiver opened his parachute and his speed changed. After a time, he reached a terminal velocity.

Explain how the forces on the skydiver caused his acceleration to vary after his parachute opened.

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(Total for question = 15 marks)

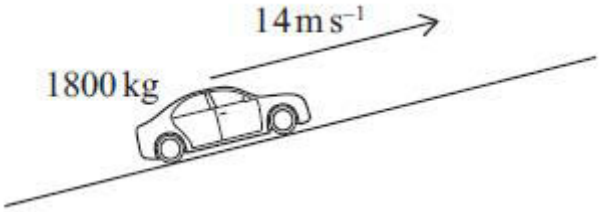
Q55.

The brakes in a car reduce the kinetic energy of the car. In an electric car, some of this energy is transferred to the battery in the car.

An electric car of mass 1800 kg moves at a speed of 14 m s^{-1} up a hill, as shown.

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The driver applies the brakes to bring the car to a stop. During braking, the vertical displacement of the car is 0.76 m and the energy stored in the battery increases by 45 kJ.

Calculate the efficiency of using the brakes to charge the battery.

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Efficiency = ...

(Total for question = 4 marks)

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